Centre For Cybersecurity Institute

Network Remote Control and Monitoring

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Create automation that would let cyber units run script to identify exposed SSH services, validate access controls under controlled conditions, and execute predefined verification tasks without manual interaction.

Project Structure:

1. User Input and Validation

1.1 Prompt the user for an IP range or subnet to scan.

1.2 Validate that the provided range is correctly formatted.

2. Scanning for SSH Services

2.1 Scan the validated IP range specifically for hosts with SSH running.

2.2 Collect and list all IPs that have an active SSH service.

3. Credential Brute Forcing

3.1 Allow the tool to use either a built-in list of SSH credentials or accept a user-provided credentials file.

3.2 Attempt to brute force SSH logins on the discovered hosts using these credentials.

4. Post-Exploitation Automation

4.1 On successful login, run a predefined command on the remote machine non-interactively (for example, create a hidden file as a proof of concept).

4.2 Do not open an interactive shell; all actions should be automated commands.

5. Automation Considerations

5.1 Handle SSH password passing automatically without interactive prompts.

5.2 Bypass host key verification prompts to ensure full automation.

5.3 All tools required for this project should be checked if it's installed on the device, or else, let the user choose to install or exit the program.

6. Output and Reporting

6.1 At the end of the scan and brute force process, generate a report of which IPs were successfully accessed and where the command was executed.

7. Creativity

Each student must add at least one improvement that they personally thought of. This can be related to design, functionality, automation flow, security, error handling, reporting, usability, or code structure. There is no required type of creativity. This addition has to be original, intentional, clearly implemented, explained and documented.

Report Format:

The report should include at least the following sections and written in formal language:

- Title page
    - The title of the project, your name, student code, the unit code, trainer name
- Table of contents
- Introduction
    - The Introduction section gives a summary of the full report, places your research in context, and emphasizes its importance. It should contain details on the report's purpose, aim, and what will be covered in the rest of the report.
- Methodologies and Discussion
    - The methodology section describes and explains what you do, with MAJOR EMPHASIS of what did not work and what are the outcomes of the various portions of the project.
        - **For each step, what didn't work?**
        - **What incorrect or no results did you get?**
        - What should be the correct method?
        - What did you learn from this?
        - Final result
- Conclusion
    - The Conclusion section restates the issues stated in the body sections and connects them to construct an argument, consolidates your evaluation of your study and data, highlighting any gaps or issues that will be addressed in the Recommendations section.
- Recommendations
    - The Recommendation section allows you to offer suggestions or solutions to the issue(s) raised in the body and/or conclusion of your report.
- References
    - Both a reference list and the proper in-text citations should be included in your report.
    - Consider using IEEE citation.